

Conditional and Independent-Event Probability

Answer Key

Precalculus / Statistics

Quick Answers

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|---|--|
| 1. $\frac{1}{3}$ | 7. 0.1846 |
| 2. $\frac{9}{20}$ | 8. (a) the expansion $= p^2 + 2pq + q^2$, — |
| 3. $\frac{3}{10}$ | 9. (a) $P(\text{test is positive}) = 0.121$, (b) $P(\text{has the condition — positive}) = 0.372$ |
| 4. (a) $P(\text{sport})$ times $P(\text{instrument}) = \frac{1}{6}$, (b) $P(\text{sport and instrument}) = \frac{3}{20}$, — | 10. (a) $P(\text{neither}) = \frac{19}{60}$, (b) $P(\text{no instrument — no sport}) = \frac{19}{30}$, — |
| 5. $\frac{1}{17}$ | |
| 6. 0.0016 | |
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What is verified

13 of 16 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 4, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus / Statistics

HSS-CP.A – problems 1, 3, 4, 8, 10

HSS-CP.B – problems 2, 5, 6, 7, 9

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

- 2. If they got 0.15: divided by the survey total instead of by the instrument row total
 - 5. If they got 0.0625: treated the second draw as if the first card had been replaced
 - 6. If they got 0.08: added the two probabilities instead of multiplying them
 - 7. If they got 0.2: multiplied the single-item defect rate by 5 instead of using the complement
 - 9. If they got 0.9: reported the test's sensitivity as the answer, confusing $P(\text{positive — has it})$ with $P(\text{has it — positive})$
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1. One of the 120 surveyed students is chosen at random. Find $P(\text{instrument})$.

Solution: The instrument row totals 40 students out of the grand total of 120, so $P = \frac{40}{120}$. $P = \frac{1}{3}$

2. A student is chosen from those who play an instrument. Find $P(\text{sport} \mid \text{instrument})$.

Solution: The condition replaces the pool: only the 40 instrument players count, and 18 of them play a sport, so $P = \frac{18}{40}$. $P = \frac{9}{20}$

3. A student is chosen from those who play a sport. Find $P(\text{instrument} \mid \text{sport})$.

Solution: Now the sport column is the pool: 60 students, of whom 18 play an instrument, so $P = \frac{18}{60}$. Note this is a different number from problem 2 — the order of the two events in a conditional probability matters. $P = \frac{3}{10}$

4. Test independence of “plays a sport” and “plays an instrument”. (a) $P(\text{sport}) \cdot P(\text{instrument})$. (b) $P(\text{sport and instrument})$. (c) What does the comparison say?

Solution (a): $P(\text{sport}) = \frac{60}{120} = \frac{1}{2}$ and $P(\text{instrument}) = \frac{40}{120} = \frac{1}{3}$, so the product is $\frac{1}{2} \cdot \frac{1}{3}$. $\frac{1}{6}$

Solution (b): 18 of the 120 students do both, so $P = \frac{18}{120}$. $\frac{3}{20}$

Solution (c) — instructor-judged. $\frac{1}{6} = \frac{20}{120}$ but the actual joint count is $\frac{18}{120}$. The two are not equal, so the events are *not* independent: knowing a student plays an instrument lowers the chance they also play a sport. Full credit requires the comparison and the “not independent” conclusion, stated in that direction.

5. Two cards are dealt from a 52-card deck without replacement. Find $P(\text{both hearts})$.

Solution: $P(\text{first heart}) = \frac{13}{52}$. After a heart is removed, 12 hearts remain among 51 cards, so $P(\text{second heart} \mid \text{first heart}) = \frac{12}{51}$. The multiplication rule gives $\frac{13}{52} \cdot \frac{12}{51} = \frac{1}{4} \cdot \frac{4}{17}$. $P = \frac{1}{17}$

6. Each part is defective with probability 0.04, independently. Two parts are selected. Find $P(\text{both defective})$.

Solution: The parts are independent, so the conditional factor equals the unconditional one and the rule collapses to a product: $P = (0.04)(0.04)$. $P = 0.0016$

7. Each part is good with probability 0.96, independently. Five are selected. Find $P(\text{at least one defective})$, to four decimal places.

Solution: “At least one” is the complement of “none”, and “none defective” means all five are good: $P(\text{all good}) = 0.96^5 = 0.8153726976$, so $P(\text{at least one defective}) = 1 - 0.96^5 = 0.1846273\dots$ $P \approx 0.1846$

- 8.** Two independent trials, success probability p , failure probability q . (a) Expand $(p+q)^2$.
 (b) What does each term count?

Solution (a): $(p+q)^2 = (p+q)(p+q) = p^2 + pq + qp + q^2$. $p^2 + 2pq + q^2$

Solution (b) — instructor-judged. p^2 is the probability of two successes; q^2 is the probability of two failures; $2pq$ is the probability of one of each. The factor of 2 is there because “one success and one failure” happens in two orders — success then failure, or failure then success — and the two products pq and qp are the two orderings. Full credit requires all three terms matched and the 2 explained as the count of orderings.

- 9.** Screening: 5% have the condition; the test is positive for 90% of those who do and 8% of those who do not. (a) $P(\text{positive})$. (b) $P(\text{condition} \mid \text{positive})$.

Solution (a): Split the positives into the two groups they can come from:

$$P(\text{positive}) = (0.05)(0.90) + (0.95)(0.08) = 0.045 + 0.076$$

$P \approx 0.121$

Solution (b): Of that total, only the first piece comes from people who really have the condition, so $P(\text{condition} \mid \text{positive}) = \frac{0.045}{0.121} = 0.37190 \dots$ $P \approx 0.372$

Note how far this is from the test’s 90% sensitivity: because the condition is rare, most positive results come from the large healthy group.

- 10.** Survey table again. (a) $P(\text{neither})$. (b) $P(\text{no instrument} \mid \text{no sport})$. (c) Does the condition change the chance?

Solution (a): 38 students play neither, out of 120, so $P = \frac{38}{120}$. $P = \frac{19}{60}$

Solution (b): The “no sport” column holds 60 students, 38 of whom play no instrument, so $P = \frac{38}{60}$. $P = \frac{19}{30}$

Solution (c) — instructor-judged. Unconditionally, $P(\text{no instrument}) = \frac{80}{120} = \frac{2}{3} \approx 0.667$. Conditionally on playing no sport it is $\frac{19}{30} \approx 0.633$. The two differ, so the condition does change the chance and the characteristics are not independent. Full credit requires both numbers compared and the conclusion drawn.